Cross-Channel Agreement Beats Consensus: Compositional Verification for Geometry Reasoning

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Abstract

Multimodal geometry reasoning requires jointly perceiving diagrams and performing symbolic derivations. Test-time verification methods such as majority voting assume voter errors are roughly independent, but in geometry all candidates perceive the same diagram, so perceptual biases cascade into correlated errors that produce confident but wrong consensus—repeating a vote within one channel cannot escape a shared misperception. To expose a structurally independent signal, we standardize each candidate into SKETCH, a format that decomposes reasoning into a neural pathway (natural-language claim) and a symbolic pathway (visual grounding $\rightarrow$ typed program $\rightarrow$ deterministic execution). Because these two pathways fail through different error processes, their agreement carries information that within-channel repetition cannot. Compositional Consensus Verification (CCV), our training-free verifier, gates consensus on this cross-pathway agreement—a precision filter that retains only candidates whose two pathways independently converge, making each concordant vote far more reliable than a raw majority vote. Across four geometry benchmarks, CCV reaches 81.0% accuracy (+10.4 pp over majority voting) with no external scorers or extra inference, and transfers to a second model family (+7.8 pp). Seven non-compositional baselines—including trained reward models—cluster at 70–71%, while cross-channel methods jump to 79–81%, confirming that structural diversity, not more votes, is the core verification resource.

1 Introduction

The verification bottleneck in geometry reasoning. Geometry reasoning differs from purely textual mathematical tasks in a fundamental way: it requires joint visual perception and symbolic derivation. A model must read spatial relationships from a diagram—angles, parallel lines, congruences—and then chain these percepts through multi-step logical deductions. This conjunction makes geometry a task that current vision-language models (VLMs) find difficult [Gao et al., 2023, Xia et al., 2025, Zhang et al., 2025b], and it also creates a distinctive failure mode for test-time verification. Figure 1 summarizes our proposed solution.

The standard approach for improving reasoning accuracy at test time is self-consistency [Wang et al., 2023]: sample K candidates and select the most frequent answer via majority voting (MV). A classical Condorcet justification assumes independent, sufficiently accurate voters, so that binary majority reliability improves as K grows. In geometry, this assumption breaks down in a specific and measurable way. All K candidates perceive the *same* diagram; if the model harbors a systematic perceptual bias—misidentifying a perpendicular as supplementary, or misreading an angle label—then a majority of candidates will share the same error. Votes become correlated because they reflect the same perceptual errors. Formally, when the pairwise error correlation ρ is substantial,

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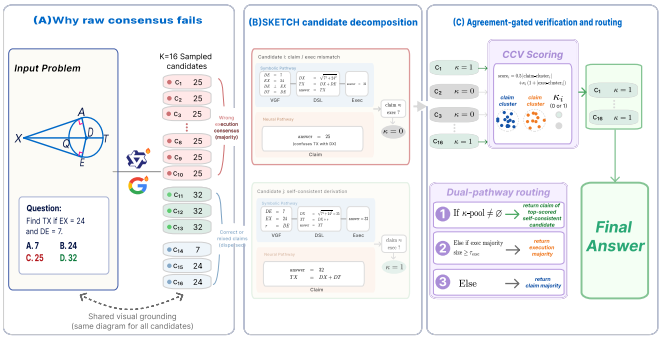


Figure 1: **CCV architecture overview.** A geometry problem is processed to sample $K{=}16$ SKETCH candidates. Each candidate exposes a symbolic reasoning pathway (visual grounding $\rightarrow$ a typed domain-specific language (DSL) $\rightarrow$ execution) and a neural intuition pathway (claim). Cross-channel agreement κ acts as a precision filter to gate consensus; when agreement collapses, the system falls back to execution or claim majority. Execution is performed externally after generation; the claim sees the DSL source but not its executed output.

the effective sample size degrades as $N_{\text{eff}} = K/(1 + (K{-}1)\rho)$ [Ladha, 1992, Nitarach, 2026]. At $\rho = 0.5$, sixteen candidates carry the information of barely two independent votes. In our experiments, wrong execution consensus clusters of size 8–16 account for 86% of the execution-first baseline failures with an oracle-correct candidate analyzed in Section 4.6—strong majorities that are confidently wrong because they derive from the same misperceived geometric facts.

Compositional structure as a verification resource. Since the problem is within-channel error correlation, a natural response is to seek a verification signal from a structurally different computation. We observe that neuro-symbolic reasoning candidates are not opaque votes. Unlike standard chain-of-thought (CoT) outputs—a single text sequence mapping to a single answer—a neuro-symbolic candidate decomposes into layers with distinct computational character:

$$\text{Candidate} = \big(\underbrace{\text{VGF}}_{\text{perception}} , \underbrace{\text{DSL}}_{\text{logic}}, \underbrace{\text{exec}}_{\text{computation}} , \underbrace{\text{claim}}_{\text{intuition}} \big) \tag{1}$$

where VGF (Visual Grounding Facts) captures geometric relations extracted from the diagram, DSL (domain-specific language) is a typed program encoding the reasoning steps, exec is its deterministic execution result, and claim is the model's stated answer.

Each candidate is generated in a single autoregressive pass in the order `[VGF]` $\rightarrow$ `[DSL]` $\rightarrow$ `[Final Answer]`; an external sandboxed interpreter executes the DSL only after generation. Thus the claim is conditioned on the DSL source, but not on its execution output. The recorded section order follows this sequence for 27,342/27,344 candidates. The claim and execution therefore share both visual premises and generated program context; their structural difference does not imply statistical independence.

Existing verification paradigms do not directly use this within-candidate comparison. *Consensus methods*, including confidence-informed self-consistency (CISC), aggregate answers within one output channel. *Learned verifiers*, including outcome reward models (ORMs), process reward models (PRMs), and their visual counterparts, train an external scorer. We instead use **compositional verification**: compare the neural claim with the externally computed program result, then use agreement to gate consensus.

Table 1: Channel coupling in the Qwen candidate pool. The within-question mean ϕ uses questions with nonconstant claim and execution correctness; failed executions ($\perp$) count as incorrect. ρ_e measures a different quantity: execution-correctness correlation between candidates. All red statistics await recomputation after removing GPS-full from the main aggregate.

Statistic	Value
Mean within-question claim–exec correctness correlation ϕ	0.325
Questions with defined ϕ	766
Within-channel execution correctness correlation ρ_e	0.526

The cross-channel verification principle. Principle. *Agreement improves the precision of retained execution votes exactly when it is more likely for correct executions than for incorrect executions.* This likelihood-ratio condition permits positively correlated channels and does not guarantee an improvement in pool-level voting accuracy (Section 3.2).

The full-pool estimate is $P(\text{exec correct} \mid \text{claim correct}) = 0.736$, above the marginal execution accuracy $p_e = 0.617$: the channels are positively correlated. Table 1 distinguishes this within-candidate coupling from correlation between candidates. Agreement can still be useful because it selects correct executions at a higher rate than incorrect ones, rather than because false agreement is necessarily uncommon.

Contributions. This paper makes the structure of neuro-symbolic candidates available for verification through SKETCH, a format exposing visual facts, a typed program, its execution, and a neural claim. The generator requires teacher-assisted reformatting and low-rank adaptation (LoRA) supervised fine-tuning (SFT), achieving >99.9% format compliance with a 9B VLM. Given an existing SKETCH pool, CCV is a training-free verifier requiring no additional model calls. Its reported gain over MV is +10.41 pp across three benchmarks, with GPS-full reported for reference; the aggregate awaits recomputation under this non-overlapping protocol. The same fixed verification protocol yields +7.84 pp on Gemma4-E4B. We characterize both recoveries and regressions, and give an exact condition for candidate-level precision improvement without assuming channel independence.

2 The SKETCH Format

Visual grounding facts (VGF). For cross-candidate verification to work, visual observations must be comparable across candidates in a structured way. Free-form natural-language descriptions of a diagram are not reliably alignable; two candidates may describe the same geometric relationship with different wording. We therefore represent visual grounding as a list of atomic facts drawn from a closed vocabulary of relations:

$$f = (\texttt{relation}, \texttt{entities}),$$
$$\texttt{relation} \in \{\texttt{parallel}, \texttt{perpendicular}, \texttt{collinear}, \ldots\} \tag{2}$$

The closed vocabulary ensures parseability and enables direct cross-candidate comparison of perceived geometric structure.

Typed DSL. The DSL is a restricted Python subset that derives the answer from VGF-grounded premises using geometry helper functions and standard mathematical operations ($\sqrt{\cdot}$, trigonometric functions, π, etc.). Restricting the subset—no imports, no I/O, no loops—ensures termination and deterministic execution, which is essential for the symbolic pathway to produce a reliable verification signal. Every program terminates with an explicit `answer = ...` assignment, so the execution result can be extracted automatically without answer parsing.

Generation and external execution. The model emits visual facts, the DSL source, and its final claim sequentially in one decode. The interpreter then executes the completed program outside the model. The resulting claim–execution comparison closes a verification loop after generation; it is not a second independently generated solution. Compositional consistency ($\kappa = 1$) supplies a useful precision filter when agreement is more likely on correct executions, as formalized below.

Training. We fine-tune Qwen3.5-9B with LoRA (r=8, α=16) on 2,012 geometry examples derived from the Geometry3K [Lu et al., 2021] training split. We use the geometry3k subset of NVIDIA Nemotron-Image-Training-v3 [Blakeman et al., 2025], which provides 2,078 CoT annotations generated by Qwen3-VL-235B. Each CoT trace is reformatted into VGF+DSL structure by DeepSeek V4 Pro. The resulting model achieves format compliance >99.9%, VGF parse success >98.7%, and DSL execution success >94.5% on test data.

3 Compositional Consensus Verification

3.1 Verification Signal Taxonomy

Given K candidates $\{c_1, \ldots, c_K\}$, each decomposed as $c_i = (\text{VGF}_i, \text{DSL}_i, \text{exec}_i, \text{claim}_i)$, we define a hierarchy of verification signals ordered by the structural diversity they require:

Level 1: Within-layer consensus. Signals computed within a single output layer, aggregated across candidates:

$$S_{\text{claim}}(z) = |\{i : \text{claim}_i = z\}| \qquad \text{(surface consensus)} \tag{3}$$

$$S_{\text{exec}}(z) = |\{i : \text{exec}_i = z,\ \text{exec}_i \neq \perp\}| \qquad \text{(execution consensus)} \tag{4}$$

Traditional MV selects $\arg\max_z S_{\text{claim}}(z)$. Execution MV selects $\arg\max_z S_{\text{exec}}(z)$.

Level 2: Cross-layer agreement. Signals that compare outputs across structurally different layers. The core signal is compositional consistency:

$$\kappa_i = \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp], \tag{5}$$

which checks whether a single candidate's neural and symbolic pathways agree. Additional cross-layer features (e.g., cross-candidate cross-layer agreement μ_i) appear in the learned ablation (Section 4.5) but are not part of the hand-designed algorithm.

Level 3: Agreement-gated consensus. Cross-layer agreement gates within-layer consensus:

$$G_\kappa(z) = \sum_i \mathbf{1}[\text{exec}_i = z] \cdot \kappa_i \tag{6}$$

Here only execution votes from self-consistent candidates count. CCV uses a related candidate-level gate: it restricts selection to self-consistent candidates and scores their claim and execution cluster support (Algorithm 1). Its execution-cluster size counts the full pool; it is not identical to G_κ.

3.2 The Cross-Channel Verification Advantage

For clarity, this analysis writes agreement as equality after answer normalization. The implementation uses approximate numeric matching and excludes failed executions from agreement. All probabilities refer to the same candidate distribution; no independence between channels or between candidates is assumed in the following precision results.

Theorem 1 (Agreement precision identity)**.** *Let X_i be the claim, Y_i the execution result, and z^* the correct answer. Define $\kappa_i = \mathbf{1}[X_i = Y_i,\ Y_i \neq \perp]$, $t_\kappa = P(\kappa_i = 1, Y_i = z^*)$, and $f_\kappa = P(\kappa_i = 1, Y_i \neq z^*)$. If $P(\kappa_i = 1) > 0$, then*

$$p_\kappa \triangleq P(Y_i = z^* \mid \kappa_i = 1) = \frac{t_\kappa}{t_\kappa + f_\kappa}. \tag{7}$$

Proposition 1 (Likelihood-ratio condition)**.** *Let $p_e = P(Y_i = z^*)$ with $0 < p_e < 1$ and $P(\kappa_i = 1) > 0$. Write $a = P(\kappa_i = 1 \mid Y_i = z^*)$ and $b = P(\kappa_i = 1 \mid Y_i \neq z^*)$. Then*

$$p_\kappa > p_e \quad \Longleftrightarrow \quad a > b. \tag{8}$$

For $b > 0$, the exact posterior-odds identity is

$$\frac{p_\kappa}{1 - p_\kappa} = \frac{p_e}{1 - p_e} \frac{a}{b}. \tag{9}$$

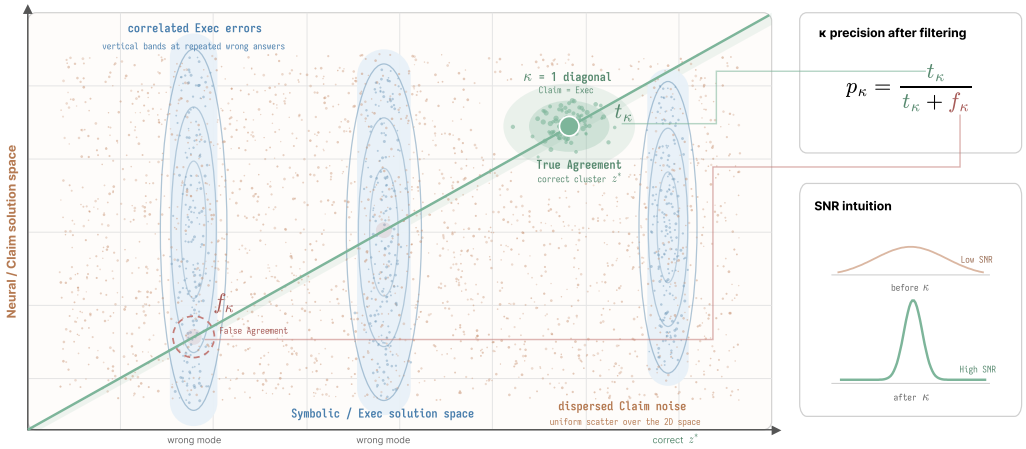


Figure 2: **Cross-channel precision filtering.** Left: κ retains claim–execution agreement, which may be either true or false. Precision improves when correct executions are retained more often than incorrect ones, without an answer-dispersion assumption. Right: in the original aggregate, higher retained precision accompanies higher correlation and a $4.1\times$ SNR-proxy improvement. Embedded statistics await the same aggregate recomputation as the red text.

This condition allows substantial false agreement. On Gemma4-E4B, 40.2% of retained candidates are falsely concordant, yet retention raises execution precision from 33.4% to 59.8%. The diagnostic is the relative retention rate for correct versus incorrect executions, not an assumption that incorrect answers are dispersed. Appendix J gives the proofs and measured likelihood ratios.

Theorem 2 (One-vs-rest SNR approximation)**.** *For n exchangeable binary correctness indicators with mean p and pairwise correlation ρ, a positive variance gives the signal-to-noise ratio (SNR)*

$$\mathrm{SNR}(p, n, \rho) = \frac{p - \frac{1}{2}}{\sqrt{p(1-p)(1 + (n-1)\rho)/n}}. \tag{10}$$

Using this quantity to compare voting reliability is a one-vs-rest approximation. A higher filtered-pool SNR requires the precision improvement to offset changes in retained pool size and correlation; the comparison and its assumptions are given in Appendix J.

Candidate precision does not imply a pool-level voting gain. Filtering can leave too few candidates, preserve correlated wrong answers, or change which answer wins a multiclass vote. Neither Proposition 1 nor the SNR approximation guarantees that CCV improves on a given question or candidate pool. The SNR also does not model CCV's weighted scoring and fallback branches.

Empirical diagnostic. In the original aggregate of 1,709 questions with 16 candidates each, execution correctness correlation is $\rho_e = 0.526$ and retained-pool correlation is $\rho_\kappa = 0.646$. Candidate precision increases from 61.7% to 87.1%, while the SNR proxy rises from 0.32 to 1.33 ($4.1\times$). These statistics illustrate a precision gain despite increased correlation; the pool-level gains are empirical results evaluated separately in Section 4. All aggregate statistics in red retain their original values pending removal of the overlapping GPS-full reference set.

A deployment pretest. Before using κ filtering in a new setting, estimate a/b on a labeled candidate pool and inspect question-stratified results. A ratio near one predicts little candidate-level precision benefit; a ratio above one supports precision filtering but still requires a separate pool-level evaluation.

3.3 κ-gated Dual-Pathway CCV

Algorithm 1 summarizes CCV. It uses claim-execution agreement as a precision gate, conceptually related to program-based consistency filtering [Toh et al., 2024], but differs by combining agreement-gated consensus with output-pathway routing. The algorithm applies cross-channel agreement as a gate at two levels:

Algorithm 1 κ-gated Dual-Pathway CCV

Require: Candidates $\{c_1, \ldots, c_K\}$, threshold τ_{exec}

1: **for** each candidate c_i **do**
2: $\quad \kappa_i \leftarrow \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp]$
3: $\quad \text{score}_i \leftarrow 0.5 \cdot |\text{claim_cluster}_i| + \kappa_i \cdot (1 + |\text{exec_cluster}_i|)$
4: **end for**
5: $\mathcal{C}_\kappa \leftarrow \{c_i : \kappa_i = 1\}$ ▷ self-consistent candidate pool
6: **if** $\mathcal{C}_\kappa \neq \emptyset$ **then**
7: $\quad$ **return** $\text{claim}(\arg\max_{c_i \in \mathcal{C}_\kappa} \text{score}_i)$ ▷ *Level 1: output neural claim*
8: **else**
9: $\quad z_{\text{exec}} \leftarrow \text{mode}(\{\text{exec}_i : \text{exec}_i \neq \perp\})$
10: $\quad$ **if** $|\{i : \text{exec}_i = z_{\text{exec}}\}| \geq \tau_{\text{exec}}$ **then**
11: $\quad\quad$ **return** z_{exec} ▷ *Level 2: fall back to symbolic execution*
12: $\quad$ **else**
13: $\quad\quad$ **return** $\text{mode}(\{\text{claim}_i\})$ ▷ *Level 3: fall back to claim majority*
14: $\quad$ **end if**
15: **end if**

Candidate scoring (Lines 2–3). The term $\kappa_i \cdot |\text{exec_cluster}_i|$ realizes agreement-gated consensus: execution consensus is trusted only when accompanied by cross-layer agreement. The term $0.5 \cdot |\text{claim_cluster}|$ provides a baseline surface signal. Raw execution cluster size receives zero weight—our learned ablation in Section 4.5 confirms this design choice.

Output pathway routing (Lines 6–12). When self-consistent candidates exist ($\mathcal{C}_\kappa$ is non-empty), the algorithm outputs the neural claim. When cross-layer agreement collapses globally, it falls back to symbolic execution consensus, but only if the execution cluster is sufficiently large ($\geq \tau_{\text{exec}}$). An empty agreement pool does not identify which channel is wrong; the execution threshold is a heuristic fallback, and the regressions in Section 4.6 show its limits.

Complexity and scope. Aggregation takes $O(K)$ time after answer normalization and clustering. On an existing SKETCH candidate pool, verification requires no additional model calls; the generator still requires LoRA SFT and teacher-assisted reformatting, so CCV is not a drop-in verifier for arbitrary unstructured pools. Aggregation costs 2–4 ms per question on one CPU core (4–7 s for the original pool); external DSL execution is shared with execution-based baselines. Appendix L separates these costs from generator training and external scoring. We select τ_{exec} on development data and examine the other fixed matching and scoring constants in Appendix M.

4 Experiments

4.1 Setup

Models and training. Our primary generator is Qwen3.5-9B, fine-tuned with LoRA (r=8, α=16) on 2,012 executor-validated Geometry3K training examples. The training data was constructed by reformatting Qwen3-VL CoT traces into the SKETCH format using DeepSeek V4 Pro. We also evaluate Gemma4-E4B to test cross-model generalization. Training was conducted using LLaMA-Factory with bf16 on 8 NVIDIA 4090 GPUs.

Evaluation protocol. Table 2 summarizes three primary benchmarks: Geometry3K test, GeoQA test, and GPS-clean, with GPS-full reported for reference. "All" aggregates only the three primary sets; heldout/OOD uses Geometry3K qid$\geq$ 300, GeoQA test, and GPS-clean, excluding both the Geometry3K dev split (qid$<$ 300) and GPS-full. Red values preserve original overlapping-pool statistics and await recomputation; they are not recomputed estimates for the revised aggregate. We use a frozen-rollout protocol with K=16 candidates, temperature 0.8, and top-p 0.95. All verification methods operate on the same rollout pool, denoted s240 for the frozen Qwen rollout pool. The execution fallback threshold τ_{exec}=9 was selected on a 300-question development split of Geometry3K and held fixed across all benchmarks and models.

Table 2: Test benchmark suite: three primary benchmarks and a reference superset. MathVista GPS denotes its geometry problem solving subset; GPS-clean is a strict subset of GPS-full. OOD denotes out-of-distribution evaluation.

Benchmark	Lang.	N	Role
Geometry3K test [Lu et al., 2021]	EN	601	In-distribution
GeoQA test [Chen et al., 2021]	ZH	754	Cross-lingual OOD
MathVista-GPS clean (GPS-c) [Lu et al., 2024]	EN	146	Cross-domain OOD
MathVista-GPS (GPS-full) [Lu et al., 2024]	EN	208	Reference (superset of GPS-clean)

GPS-clean contains 146 of the 208 GPS-full questions. GPS-full is reported separately and is excluded from the main aggregate to avoid duplicate counting.

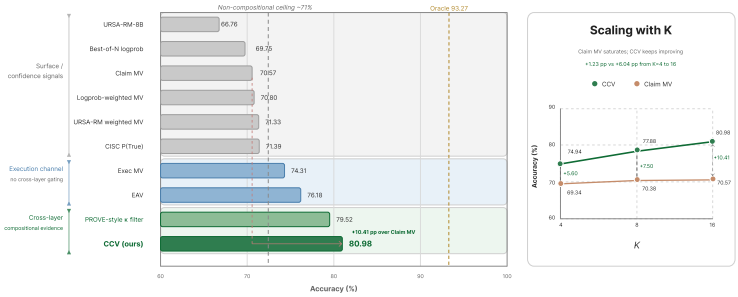


Figure 3: **Evidence ladder and K scaling.** Surface and confidence-based selectors cluster near 70–71%, execution-only methods improve moderately, and cross-layer compositional verification produces the main jump to 79–81%. As K increases, the CCV gain over Claim MV grows, indicating that compositional verification unlocks additional rollout diversity. Embedded aggregate statistics await recomputation without GPS-full.

4.2 Main Results

Execution-Answer Voting (EAV) is the execution-first priority baseline. Figure 3 visualizes the same evidence ladder and K-scaling trend, while Table 3 reports the full benchmark breakdown. The four surface and confidence-based selectors—Claim MV, logprob-weighted MV, CISC, and URSA-RM weighted MV—all lie in a narrow 70.57–71.39% band, despite spanning self-confidence weighting and an external learned PRM. The learned ablation in Section 4.5 shows a similar range for logprob-based best-of-N and learned surface+execution feature selectors. Even URSA-RM, the strongest external scorer, achieves only $+0.76$ pp over Claim MV when used for weighted voting. Execution-only methods improve to 74–76%, but the large jump appears only when cross-layer agreement is used as a compositional signal: a PROVE-style κ filter reaches 79.52%, and CCV reaches 80.98%, a $+10.41$ pp improvement over Claim MV and $+4.80$ pp over EAV.

The original paired per-question analysis over 1,709 entries reported the following gains, confidence intervals, and tests, all of which require recomputation without GPS-full: CCV improves over Claim MV by $+10.42$ pp (95% paired bootstrap CI $[8.48, 12.40]$ pp; exact McNemar $p < 10^{-12}$) and over EAV by $+4.80$ pp (95% CI $[3.39, 6.26]$ pp; exact McNemar $p = 5.7\times10^{-11}$). The original heldout analysis also reports gains away from the development split: on the disjoint heldout/OOD split (excluding dev 300), CCV reaches 81.93%. Finally, a sweep over the development-selected threshold, τ_{exec}, reports a broad plateau: across $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies by 1.00 pp and heldout/OOD accuracy by 1.06 pp (Appendix C). The κ matching threshold itself is also stable: varying the candidate-level relative tolerance around the default leaves CCV accuracy unchanged on the full s240 pool (Appendix H).

4.3 Cross-Model Generalization

To test whether CCV's gains depend on the specific base model, we repeat the full evaluation protocol with Gemma4-E4B, a 4B-parameter multimodal model with substantially weaker geometry reasoning capacity. The Gemma4 model achieves comparable SKETCH format compliance to the

Table 3: Evidence ladder across verification signal types (K=16). All aggregates G3K, GeoQA, and GPS-c only; GPS (GPS-full), separated on the right, is a reference superset excluded from All. Red All values await recomputation. Bold marks the best non-oracle method in each column; PROVE-style filtering leads on GeoQA. Surface/confidence methods, including learned PRM scores and self-judged confidence, cluster near 70–71%. Execution-only signals improve moderately. Cross-layer compositional verification produces the large jump to 79–81%.

Method	Type	G3K	GeoQA	GPS-c	All	GPS
Surface / confidence signals						
Claim MV	consensus	66.72	73.61	72.60	70.57	69.23
Logprob-weighted MV	weighted consensus	66.56	73.87	72.60	70.80	70.67
CISC $P(\text{True})$-weighted	confidence consensus	68.05	74.27	72.60	71.39	69.71
URSA-RM weighted MV	learned verifier	62.73	75.60	79.45	71.33	75.00
Execution channel without cross-layer gating						
Exec MV	exec consensus	72.71	75.60	76.71	74.31	72.60
EAV	exec-first priority	74.04	77.72	77.40	76.18	75.96
Cross-layer compositional signals						
PROVE-style κ filter	binary agreement	76.54	**82.10**	81.51	79.52	77.40
CCV (ours)	gated + routing	**78.54**	81.83	**86.30**	**80.98**	**81.25**
Oracle	upper bound	93.51	93.50	91.78	93.27	92.79

Qwen model, so this experiment is a confirmation of the verification mechanism rather than a new generator contribution.

Table E.1 reports the cross-model evidence ladder using the same τ_{exec}=9 for both models. First, the absolute accuracy gap is large: Claim MV drops from 70.57% on Qwen3.5-9B to 44.94% on Gemma4-E4B, consistent with the capacity difference. Second, the CCV gain pattern is preserved: CCV improves over Claim MV by +10.41 pp on Qwen and +7.84 pp on Gemma4. The Gemma4 result is especially diagnostic because its execution pathway alone is worse than the claim pathway (Exec MV 41.08% vs Claim MV 44.94%), making EAV ineffective (44.88%). Yet CCV still reaches 52.78%, showing that it exploits structural cross-channel agreement rather than the raw accuracy advantage of execution votes.

The precision-filtering pattern also replicates quantitatively: κ-filtering yields nearly identical precision gains across models (+25.40 pp vs +26.38 pp), and the counterintuitive correlation pattern repeats ($\rho_\kappa > \rho_e$ while SNR improves; Table D.1). This supports transfer between the two tested model families; it does not establish model-independent gains. The CCV advantage also scales with K on both Qwen and Gemma4 (Appendix D, Appendix F).

4.4 K Scaling Analysis

Figure 3 shows that the CCV advantage scales with K: the gain over Claim MV grows from +5.60 pp at K=4 to +10.41 pp at K=16 (tabulated in Appendix D). Surface consensus saturates quickly (Claim MV gains only +1.23 pp from K=4 to 16), while compositional verification continues extracting signal from larger candidate pools.

As a negative control, VGF consensus—another within-layer signal—not only fails to help but *hurts* performance when added (from 80.05% to 79.93%). This reinforces the central finding: within-layer consensus—whether the layer is claim, execution, or VGF—is vulnerable to correlated perception errors. In these comparisons, the substantial gains come from cross-layer gating.

4.5 Learned Feature Ablation

To validate that the compositional features carry the transferable verification signal—not just that our specific CCV rule is effective—we train candidate-level logistic regression classifiers on Geometry3K dev (300 questions) and evaluate on disjoint OOD benchmarks.

Table D.3 reports the full ablation. In this ablation, the substantial heldout gains come from compositional features: logprob best-of-N, surface features, and surface+execution features re-

main near Claim MV (71.11–71.86% OOD), while adding cross-layer features jumps to 79.85% OOD and C-only features reach 81.02%. The learned model rediscovers the gating principle: raw exec_cluster_size has a negative coefficient (-2.40), while $\kappa \times$ exec_cluster is strongly positive ($+1.96$). A simple rule is therefore enough: PROVE-style κ filtering reaches 81.02% OOD, and dual-output CCV reaches 81.93% OOD.

4.6 Recoveries and Regressions

CCV changes answers in both directions. Relative to Claim MV, it records 237 wins and 59 losses (4.02:1); relative to EAV, it records 121 wins and 39 losses (3.10:1). The per-benchmark net gains over Claim MV are positive, ranging from $+8.22$ pp on GeoQA to $+13.70$ pp on GPS-clean. Appendix K reports both 2×2 exchange tables and the regression categories.

The recovery-only taxonomy considers 355 EAV failures with an oracle-correct candidate. Wrong execution consensus accounts for 305/355 (86%). For the Exec✗ Claim✓ subtype, where the execution majority is wrong but the claim majority is correct, CCV recovers 123/139 (88.5%). The Exec✗ Claim✗ subtype remains difficult (8.3% recovered). These subtype counts and the paired exchange tables come from different reported analyses and must be reconciled in the aggregate audit; the flow figure describes recoveries, not net gains. Among self-consistent candidates, reliability rises from 19% at execution cluster size 1 to 93% at size 8+, supporting the combination of agreement and consensus.

Regressions expose three limits. False κ agreement explains 26/59 losses against Claim MV and 30/39 against EAV, often with small wrong agreement groups (median sizes 3 and 2, respectively). Incorrect fallback execution consensus explains 33 losses against Claim MV: none of their candidates executes to the correct answer, and 15 have unanimous 16/16 execution consensus. Threshold effects account for 7 losses against EAV. Agreement and strong fallback consensus can both preserve shared errors.

5 Related Work

Geometry reasoning systems mainly improve the generator side of the problem. Specialized VLMs such as G-LLaVA [Gao et al., 2023], GeoX [Xia et al., 2025], and MAVIS [Zhang et al., 2025b] strengthen multimodal geometry solving on benchmarks such as Geometry3K [Lu et al., 2021] and GeoQA [Chen et al., 2021]. Neuro-symbolic solvers such as AlphaGeometry [Trinh et al., 2024, Chervonyi et al., 2025] and FormalGeo [Zhang et al., 2023] use symbolic structure to derive proofs or answers. Our work is complementary: it assumes a candidate pool already exists and studies how to verify candidates when shared diagram perception makes their errors correlated.

Test-time verification is usually based on either consensus or learned scoring. Self-consistency [Wang et al., 2023], universal self-consistency [Chen et al., 2023], and confidence-weighted CISC [Taubenfeld et al., 2025] aggregate opaque answer votes, which works best when sampled errors are sufficiently independent. Learned verifiers, including ORMs/PRMs [Cobbe et al., 2021, Uesato et al., 2022, Lightman et al., 2024, Wang et al., 2024], GenRM [Zhang et al., 2025a], URSA-RM [Luo et al., 2025], and VisualPRM [Wang et al., 2025], train external scorers and add verifier-specific data or inference. CCV differs from both paradigms by deriving verification signal from structure already present inside each candidate.

The closest line of work uses programs or error-correlation analysis to diagnose voting failures. PROVE [Toh et al., 2024] filters candidates whose translated Python output matches the natural-language answer, a binary consistency signal analogous to κ. Our framework generalizes this idea to geometry-specific neuro-symbolic outputs, where VGF, typed DSL, deterministic execution, and neural claims form a hierarchy of cross-layer signals. Nitarach [2026] show that correlated errors limit majority voting; CCV targets this regime directly by gating consensus through cross-channel agreement rather than trying to decorrelate samples.

6 Limitations

Coupled generation. The claim and DSL are produced by one autoregressive model, and the claim conditions on the DSL source. Their mean within-question correctness correlation is $\phi =$

0.325; they are not independent solutions. The likelihood-ratio condition characterizes retained-candidate precision, while pool-level selection gains remain empirical and can fail under correlated false agreement.

Generator and teacher dependence. The training-free verifier operates only once a SKETCH pool exists. Our generators require LoRA SFT on teacher-assisted reformatted traces; this preparation is part of the method's cost and limits applicability to arbitrary existing candidate pools. Closed-source models might emit SKETCH through few-shot prompting, but we have not tested this and make no performance claim for them.

Format degradation and fallback. Malformed or non-executable programs cannot contribute execution votes and are excluded from κ agreement. Format degradation therefore shrinks the agreement pool and can trigger fallback, rather than silently treating a failed execution as concordant. This handling does not make fallback reliable: well-formed programs can execute incorrect premises, and strong execution consensus can still be wrong.

Task and domain boundaries. We validate numeric and multiple-choice geometry reasoning on two open-weight model families. Transfer to chart/table question answering, physics diagrams, or code generation is a hypothesis that should first be tested by estimating the agreement likelihood ratio on labeled pools. A ratio near one predicts little precision benefit. Open-ended geometry proofs require agreement at the level of proof-checker judgments, replacing numeric equality; that is a method extension, not a change of tolerance.

Selection-only headroom. Only 8.3% of the reported Exec✗ Claim✗ subtype is recovered. Selection cannot produce an answer absent from the candidate pool: reported oracle accuracies are 93.27% for Qwen and 91.75% for Gemma. This separates remaining error into cases where no correct candidate is available and cases where a correct candidate exists but the selector misses it. Premise verification or improved generation is needed to address the first component; the exchange tables characterize risks in the second.

7 Conclusion

SKETCH exposes a claim and an externally executed program result for each geometry candidate. CCV is a training-free verifier over these structured pools, with no additional model calls at verification time. Its precision benefit requires agreement to be more likely for correct executions than for incorrect ones, a condition compatible with positively correlated channels and substantial false agreement. The reported geometry comparisons place non-compositional methods around 70–71% and cross-channel methods around 79–81%, with both recoveries and regressions; aggregate values await recomputation on three benchmarks, with GPS-full reported for reference. Benefits beyond geometry remain a hypothesis. A falsifiable next step is to measure the likelihood ratio on labeled chart/table QA, physics-diagram, and code-generation pools before testing pool-level selection gains.

Planned release. We will release evaluation and aggregation scripts, frozen candidate pools and per-question selections, external scorer files, and audit scripts to support reproducibility of the verification comparisons.

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A Training Data Construction

Table A.1 summarizes the full construction pipeline for the 2,012-example SFT set. We start from the official Geometry3K [Lu et al., 2021] training split, use the geometry3k subset of Nemotron-Image-Training-v3 [Blakeman et al., 2025] as the Qwen3-VL-235B-A22B-Thinking CoT source, reformat each matched trace into the SKETCH VGF+DSL+answer format with DeepSeek V4 Pro, and retain only examples whose DSL executes and matches the official answer. The reformatting stage is text-only: the expensive visual reasoning is supplied by the Qwen3-VL CoT trace, while the official gold answer is used as the final target for machine-checkable reformatting.

Table A.1: Training data construction and filtering. Percentages are relative to the 2,078 matched CoT rows unless otherwise noted.

Stage	Count	Rate
Official Geometry3K train rows	2,101	–
Matched Qwen3-VL CoT traces	2,078	98.91% of official
Qwen3-VL boxed-answer correct	2,065	99.37%
DeepSeek V4 Pro reformat attempts	2,078	100.00%
Strict validator pass	1,963	94.47%
Recovered by tolerant VGF normalization	49	2.36%
Final executor-validated SFT rows	2,012	96.82%
Filtered: execution-answer mismatch	61	2.94%
Filtered: DSL execution/safety error	5	0.24%

The final 66 filtered examples consist of 61 execution-answer mismatches and five DSL execution/safety failures: one undefined `UB`, one undefined `z`, one undefined `m`, and two blocked-token errors. The tolerant VGF pass clears unsupported or over-arity visual facts only when the DSL and answer remain executor-valid; a final post-processing pass keeps 2,012 rows, with 41 DSL snippets changed by the sanitizer and 6 VGF lists cleared.

B Data Construction Prompts

Stage 1: stored source prompt for frontier CoT traces. We did not rerun Qwen3-VL-235B-A22B-Thinking for this paper. We use an existing CoT source whose stored user messages follow the problem image plus the following prompt pattern:

```
Solve this geometry problem. Study the diagram and select
the correct answer.

{problem_text}
{choices}

Do not explain your answer but write only the answer label
(A, B, ...) in this format:
Answer: \boxed{...}.
```

Stage 2: VGF+DSL reformatting. DeepSeek V4 Pro receives no image input. It reformats the official problem, choices, official answer, and Qwen3-VL CoT into the target SKETCH format using the following template:

```
Convert the provided Qwen3-VL-235B natural-language geometry
chain-of-thought into the target VGF+DSL format.
Do not solve from scratch. Use the teacher CoT as the visual
and reasoning evidence, and use the official gold answer as
the final target.
Return exactly three sections in this order and no extra sections:
[VGF]
[DSL]
[Final Answer]

[VGF] rules:
Output a raw JSON list only. Do not use markdown fences.
Each item must be {"relation": <relation>,
"entities": [<tokens>]} with exact relation arity.
Allowed relations: {allowed_vgf_relations}.
Relation arities: {relation_arities}.
For unary shape relations, put the whole object in one entity
token, e.g.
{"relation":"parallelogram","entities":["MNPQ"]}.
Use at most 4 facts to stay safely under the validator limit
of 5 facts.
Use [] if no reliable visual fact is needed.

[DSL] rules:
Output executable Python assignment statements only.
Do not use markdown fences.
Do not include comments or prose.
Never write import statements, including 'from math import ...'.
Math functions are already available by name.
Every variable must be assigned before it is used.
Do not write placeholder equations with unknown symbols.
The final assignment must define a variable named answer.
The value of answer must evaluate to the official gold option
value; use the teacher CoT equations to derive that value.
Allowed math names: sqrt, pi, sin, cos, tan, radians, degrees,
asin, acos, atan, abs, min, max, round.
Do not read files, import modules, call external tools, or use
input/output functions.
[Final Answer] must be exactly the official gold option letter:
{gold}.
```

```
In [DSL], set answer to the numeric or symbolic value of
option {gold}:
{gold_value}.

Official problem:
{problem_text}
{choices}

Official gold answer: {gold}
Official gold option value: {gold_value}
Teacher boxed answer: {qwen_boxed_answer}

Teacher NL CoT:
{qwen_cot}
```

C τ_{exec} Sensitivity

The development-selected parameter in CCV is the execution-majority threshold τ_{exec}, used only when the κ-pool is empty. We sweep $\tau_{\text{exec}} = 1 \ldots 16$ on the main Qwen3.5-9B s240 K=16 rollout pool, using the same split protocol as the main paper: Geometry3K qid<300 for dev, and Geometry3K qid≥ 300 plus GeoQA and GPS-clean for heldout/OOD evaluation; GPS-full is excluded. Figure C.1 plots the full sweep, and Table C.1 reports selected points. We use the resulting τ_{exec}=9 for both Qwen3.5-9B and Gemma4-E4B.

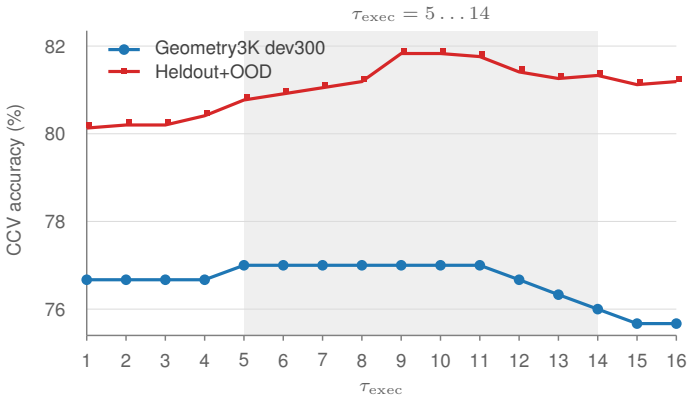


Figure C.1: **Sensitivity to τ_{exec}.** CCV is stable over a broad range of execution thresholds. In the plateau $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies from 76.00% to 77.00%, and heldout/OOD accuracy varies from 80.77% to 81.83%. Embedded curve statistics await recomputation without GPS-full.

Table C.1: Selected points from the τ_{exec} sweep (full curve in the figure above). The dev-selected threshold is $\tau_{\text{exec}} = 9$.

τ_{exec}	Dev@300	Heldout/OOD	All
1 (no threshold)	76.67	80.13	79.52
9 (selected)	**77.00**	**81.83**	**80.98**
16 (maximum)	75.67	81.19	80.22

D Additional Diagnostics

This appendix contains the diagnostic tables and flow visualization referenced by the main experimental analysis. Table D.1 reports the precision-filtering statistics behind the cross-model claim, Table D.2 gives the exact Qwen K-scaling values summarized in Figure 3, Table D.3 gives the learned feature ablation, and Figure D.1 visualizes the error taxonomy and routing behavior.

Table D.1: Cross-model diagnostic for the precision-filtering mechanism. The κ-filtered pool is more correlated than the raw execution pool, but its much higher per-candidate precision improves the effective verification signal.

Model	p_e (%)	p_κ (%)	Δp (pp)	ρ_e	ρ_κ	mean K_κ	SNR raw$\rightarrow\kappa$
Qwen3.5-9B	61.70	87.10	+25.40	0.526	0.646	9.09	0.32$\rightarrow$1.33
Gemma4-E4B	33.43	59.81	+26.38	0.504	0.630	5.77	$-$0.48$\rightarrow$0.24

Table D.2: Performance across K for the three primary benchmarks, with GPS-full reported separately for reference. The original aggregate values shown here await recomputation excluding GPS-full.

K	Claim MV	Exec MV	CCV	Oracle	CCV $-$ Claim MV
4	69.34	68.98	74.94	84.74	+5.60
8	70.38	72.43	77.88	89.89	+7.50
16	70.57	74.31	80.98	93.27	+10.41

E Cross-Model Validation

Table E.1 evaluates the same verification protocol on a lower-capacity Gemma4-E4B generator, using the same τ_{exec}=9 as Qwen3.5-9B. The Gemma4 pool uses the same frozen K=16 protocol as Qwen: Geometry3K test (601), GeoQA test (754), and GPS-clean (146) form the three primary benchmarks, with GPS-full (208) reported for reference. The original overlapping aggregate contained 1,709 entries and 27,344 candidates; all aggregate results below await recomputation excluding GPS-full. The evaluation uses the standard benchmark files; it does not use the deprecated 400-question checkpoint-selection set.

The notable difference is that Gemma4 reverses the raw pathway ordering: Exec MV is below Claim MV (41.08% vs 44.94%), whereas Qwen's execution pathway is stronger than its claim pathway. Consequently, EAV, an execution-first priority rule, is ineffective on Gemma4 (44.88% vs 44.94% Claim MV). One possible explanation is weaker visual grounding: within-layer execution consensus is more easily dominated by correlated perception errors. The observed gain does not require execution to be the stronger raw channel. CCV does not trust the raw execution vote; it uses claim–execution agreement as a precision filter. Even when the execution pathway is worse in aggregate, candidates where claim and execution agree remain informative, yielding +7.84 pp over Claim MV and +7.90 pp over EAV on Gemma4.

F Gemma4 K Scaling

Table F.1 repeats the K-scaling analysis on the frozen Gemma4-E4B K=16 pool by subsampling candidates; $K<16$ rows are averaged over four random subsamples. We keep τ_{exec}=9 fixed to match the cross-model comparison in the main text.

The Gemma4 gain increases with candidate budget, from +4.87 pp at K=4 to +7.84 pp at K=16, matching the qualitative scaling trend observed for Qwen.

G Gemma4 Implementation Details

The Gemma4-E4B experiment uses the frozen checkpoint-720 LoRA adapter and the frozen rollout pool described in Appendix E; no additional model calls are made for the verification comparisons. The adapter uses LoRA rank 8 and $\alpha = 16$. On a 400-example format audit, the model reaches 96.0% complete-section rate, 99.0% VGF parse success, 96.0% DSL syntax success, and 94.25% DSL execution success, comparable to the format reliability of the Qwen model.

Table D.3: Compositional feature ablation. "S" = standard features (logprob, claim cluster), "E" = execution features, "C" = compositional cross-layer features (κ, $\kappa \times$ exec_cluster, μ, $\kappa \times \mu$, sc_rate). The last column reports OOD gain over Claim MV.

Method / Features	Dev@300	OOD	All	Δ OOD vs Claim MV
Claim MV (reference)	66.67	71.77	70.57	0.00
Best-of-N (logprob)	66.00	71.11	69.75	−0.67
LogReg S-only	67.00	71.86	70.74	+0.08
LogReg S+E	66.33	71.69	70.63	−0.08
LogReg S+E+C	74.67	79.85	78.47	+8.08
LogReg C-only	75.33	81.02	79.64	+9.24
PROVE-style κ filter (rule)	75.00	81.02	79.52	+9.24
Dual-output CCV (ours, rule)	77.00	81.93	80.98	+10.16

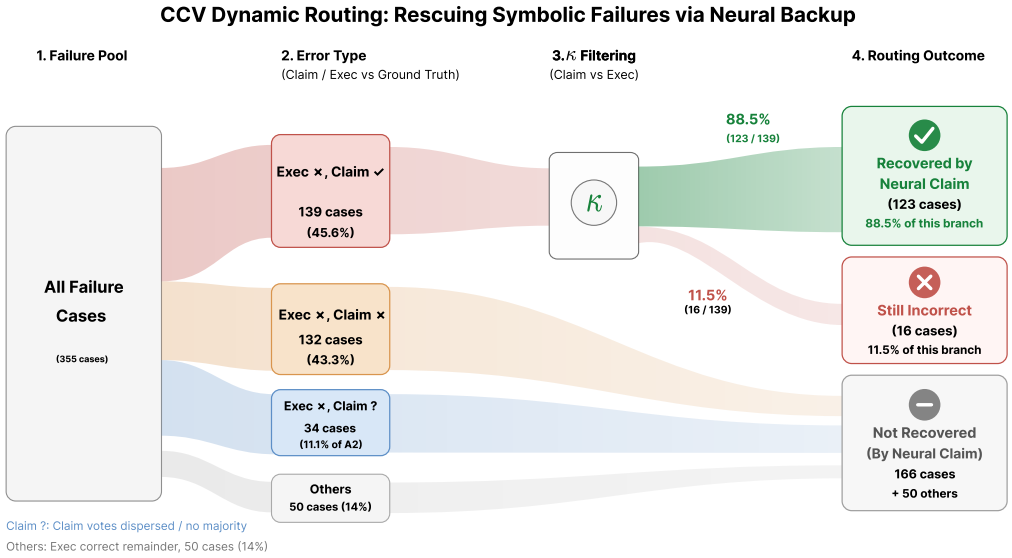


Figure D.1: **Error flow and dynamic routing.** Among 355 EAV failures with an oracle-correct candidate, wrong execution consensus dominates (86%) and splits into three subtypes: Exec✗Claim✓, Exec✗Claim✗, and Exec✗Claim?. In the recoverable Exec✗Claim✓ stream, the claim majority is correct while the execution majority is wrong; CCV's κ filter routes 123/139 cases (88.5%) to the correct neural-claim output, illustrating how output-pathway routing rescues samples when the symbolic channel collapses. This figure shows only the recovery direction; regressions are reported in Tables K.1–K.3. Embedded counts await audit against those paired tables.

H κ Matching Threshold Sensitivity

The relation $X_i \approx Y_i$ is implemented by normalizing the claimed final answer and executed DSL answer, then applying numeric tolerance. Floating-point answers, fractions, square roots, and π expressions are parsed to real values; degree symbols are treated as unit suffixes and removed before numeric comparison. For multiple-choice questions, numeric answers are first mapped to option letters using the fixed choice tolerance 0.02. Table H.1 reports an ablation that keeps the numeric absolute tolerance fixed at 10^{-3} and sweeps only the candidate-level relative tolerance used for κ.

All threshold settings yield identical results because geometry answers are effectively discrete: integers, simple fractions, radicals, angles, or multiple-choice letters. In this domain, claim–execution agreement is almost always binary: pairs either match exactly, or differ by margins much larger than even the loosest tolerance. This sweep supports stability over the tested relative tolerances on the original pool; the revised aggregate and other constants are audited separately (Appendix M).

Table E.1: Cross-model validation on the same frozen $K=16$ protocol for three primary benchmarks, with GPS-full excluded from the aggregate and reported for reference, using $\tau_{\text{exec}}=9$ for both models. Absolute accuracy differs because Qwen3.5-9B has higher model capacity than Gemma4-E4B, but the gain pattern is consistent: cross-layer κ filtering and CCV improve over surface consensus for both models with no additional model calls for verification on existing SKETCH pools. Red values are the original aggregates pending recomputation; generator SFT and teacher-assisted reformatting costs are separate (Appendix L).

Method	Qwen3.5-9B	Gemma4-E4B
Claim MV	70.57	44.94
Exec MV	74.31	41.08
EAV	76.18	44.88
PROVE-style κ filter	79.52	51.90
CCV (ours)	**80.98**	**52.78**
CCV − Claim MV	+10.41	+7.84
CCV − EAV	+4.80	+7.90
Oracle	93.27	91.75

Table F.1: Gemma4-E4B K scaling from the frozen $K=16$ pool. Entries for $K=4, 8$ are mean±std over four subsamples.

K	Claim MV	Exec MV	EAV	PROVE-style κ	CCV	Oracle	CCV − Claim MV
4	42.41±0.81	37.60±0.42	46.94±0.46	46.65±0.74	47.28±0.60	72.24±0.60	+4.87
8	44.00±0.48	39.86±0.24	48.82±0.79	50.04±0.62	50.85±0.58	83.84±0.32	+6.85
16	44.94	41.08	44.88	51.90	52.78	91.75	+7.84

I Qualitative Case Studies

We present four representative cases from the frozen Qwen3.5-9B s240 $K=16$, $T=0.8$ pool, selected to illustrate the four behavioral modes of CCV: (1) successful rescue when execution consensus is wrong but claim is correct, (2) hard cases where both pathways agree on a wrong answer, (3) correlated perception errors that produce incorrect agreement, and (4) fallback routing when no self-consistent candidate exists. Each case reports method outputs and the key DSL of the CCV-selected candidate.

I.1 CCV Rescue (Exec✗ Claim✓)

`geometry3k_test` qid 325. Expected: B (16). Choices: A=12, B=16, C=51, D=65.

Method outputs. Claim MV: B; Exec MV: C; EAV: C; CCV: **B**.

CCV-selected DSL ($\kappa = 1$, claim cluster 9, exec cluster 6):

```
angle_X = angle_L
x = angle_X / 4
answer = x          # 16.25 -> B
```

CCV selects a κ-consistent candidate whose claim and execution agree on B, while EAV follows a wrong execution majority (C). In this example the claim majority is correct and the execution majority is wrong, while the selected candidate agrees on the correct option.

I.2 Hard Case: Exec✗ Claim✗

`geometry3k_test` qid 72. Expected: A (3). Choices: A=3, B=4, C=6, D=7.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 15, exec cluster 15):

```
x = 4
answer = x          # 4.0 -> B
```

Table H.1: Sensitivity to the candidate-level relative tolerance in $X_i \approx Y_i$ on the frozen Qwen3.5-9B s240 K=16, T=0.8 pool.

Setting	Candidate relative tolerance	CCV accuracy	κ candidates / precision
10^{-6}	0.000001	80.98	15,535 / 87.07
10^{-4}	0.0001	80.98	15,535 / 87.07
10^{-3}	0.001	80.98	15,535 / 87.07
default (10^{-2})	0.01	80.98	15,535 / 87.07
5×10^{-2}	0.05	80.98	15,535 / 87.07

All methods agree on the wrong answer. This case shows that cross-channel agreement can itself be wrong; agreement alone cannot establish correct premises.

I.3 Correlated Perception Error

`geometry3k_test` qid 35. Expected: D (155). Choices: A=20, B=25, C=30, D=155.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 16, exec cluster 16):

```
arc_BC = 30
arc_AD = 20
angle_2 = (arc_BC + arc_AD) / 2
answer = angle_2    # 25.0 -> B
```

This is a boundary case for CCV: a correlated visual error makes both channels self-consistent around the wrong answer. With 16/16 candidates in both clusters, selection alone has no alternative evidence to break the incorrect consensus.

I.4 Empty κ-Pool with Execution Fallback

`geometry3k_test` qid 56. Expected: C (70). Choices: A=66, B=68, C=70, D=72.

Method outputs. Claim MV: D; Exec MV: C; EAV: C; CCV: **C**.

CCV-selected DSL ($\kappa = 0$, claim cluster 10, exec cluster 16):

```
leg1 = 24
hypotenuse = 74
x = sqrt(hypotenuse**2 - leg1**2)
answer = x          # 70.0 -> C
```

The κ-pool is empty (no candidate's claim matches its execution), so CCV falls back to the execution-majority pathway. The large exec cluster (16/16) exceeds τ_{exec}, providing the correct answer. This illustrates dual-pathway routing: when cross-channel agreement is absent, the deterministic symbolic path remains available as a fallback.

J Proofs for the Cross-Channel Verification Analysis

J.1 Definitions without an independence assumption

Let X_i be a normalized claim and Y_i a normalized execution answer, allowing $Y_i = \perp$ for failure; z^* denotes the ground-truth answer for the corresponding question. Let $E_i = \mathbf{1}[Y_i = z^*]$ and $C_i = \mathbf{1}[X_i = z^*]$. Agreement is $\kappa_i = \mathbf{1}[X_i = Y_i, Y_i \neq \perp]$. Failed executions count as incorrect and cannot agree. The precision identities hold for any joint distribution of (X_i, Y_i), including the dependence created by generating the claim after the DSL source.

Within-channel correlation $\rho_e = \mathrm{Corr}(E_i, E_j)$ summarizes different candidates from the same question; ρ_κ is its counterpart within retained candidate pairs. In contrast, the within-question ϕ coefficient correlates C_i and E_i across candidates of one question. We report its mean over questions

where both binary variables have nonzero variance. These correlations measure different dependence relationships and cannot be substituted for one another.

J.2 Precision identity and likelihood-ratio proof

Theorem 1 follows by partitioning agreement into correct and incorrect execution: $P(\kappa = 1) = t_\kappa + f_\kappa$. Bayes' rule gives $p_\kappa = t_\kappa/(t_\kappa + f_\kappa)$ whenever the denominator is positive.

For Proposition 1, let $a = P(\kappa = 1 \mid E = 1)$ and $b = P(\kappa = 1 \mid E = 0)$. Since $0 < p_e < 1$ and $P(\kappa = 1) > 0$,

$$p_\kappa = \frac{p_e a}{p_e a + (1 - p_e) b}, \tag{J.1}$$

$$p_\kappa - p_e = \frac{p_e(1 - p_e)(a - b)}{p_e a + (1 - p_e) b}, \tag{J.2}$$

$$p_\kappa > p_e \iff a > b. \tag{J.3}$$

For $b > 0$, dividing the first expression by its complement yields Eq. (9). If $b = 0$, positive retention implies $a > 0$ and $p_\kappa = 1$; if $a = b = 0$, the retained pool is empty and its precision is undefined. No conditional-independence or wrong-answer-dispersion assumption is needed.

Table J.1: Agreement diagnostics on the original pools, all pending recomputation under the three-benchmark protocol. The pooled likelihood ratio is a/b; the question-stratified Mantel–Haenszel estimate excludes failed executions and is a separate association diagnostic, not the ratio used in the pooled odds identity. Pool-level CCV gains are measured separately.

Metric	Qwen3.5-9B	Gemma4-E4B
$P(\text{agree} \mid \text{exec } \checkmark)$	0.802	0.646
$P(\text{agree} \mid \text{exec } \times)$	0.192	0.218
Pooled likelihood ratio a/b	4.18×	2.96×
Within-question Mantel–Haenszel (excluding $\perp$)	3.22×	2.09×
CCV relative to Claim MV	+10.41 pp	+7.84 pp

Table J.1 reports agreement enrichment for both model families. Before deployment on a new labeled pool, this comparison can reject settings with little expected precision benefit. Question-stratified diagnostics help distinguish within-question association from differences in question difficulty. Neither diagnostic establishes that the resulting selector will win a pool-level vote.

J.3 SNR derivation and scope of Theorem 2

For n exchangeable binary correctness indicators with common mean p and pairwise correlation ρ,

$$\operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} B_i\right) = \frac{p(1-p)}{n}\,[1 + (n-1)\rho]. \tag{J.4}$$

Dividing $p - \frac{1}{2}$ by the standard deviation gives the SNR in Theorem 2. This variance identity is exact under the stated assumptions; using SNR to approximate majority reliability requires an additional distributional approximation and does not describe multiclass answer counts exactly.

For $p_e > \frac{1}{2}$, nonempty fixed retained size K_κ, and positive variances, $\mathrm{SNR}_\kappa > \mathrm{SNR}_{\mathrm{raw}}$ is equivalent to

$$\frac{p_\kappa - \frac{1}{2}}{p_e - \frac{1}{2}} > \sqrt{\frac{p_\kappa(1 - p_\kappa)}{p_e(1 - p_e)} \cdot \frac{K(1 + (K_\kappa - 1)\rho_\kappa)}{K_\kappa(1 + (K - 1)\rho_e)}}. \tag{J.5}$$

When $p_e \le \frac{1}{2}$, compare the signed SNR expressions directly instead of dividing by $p_e - \frac{1}{2}$; in particular, the displayed inequality cannot be applied unchanged to Gemma. Substituting a mean retained size for random K_κ is another approximation. Candidate-level precision improvement alone does not establish this SNR comparison, and neither establishes a gain for CCV's weighted selection and fallback policy.

J.4 Measured diagnostic illustration

Table J.2 preserves the original post-hoc Qwen statistics. Failed executions count as incorrect; within-channel correlations use ordered candidate pairs from the same question. The retained size is a mean, so these are descriptive SNR proxies rather than predicted selection accuracies.

Table J.2: Original statistics for the one-vs-rest SNR proxy, pending recomputation.

Method	n	p	ρ	SNR
Raw exec MV	16	0.617	0.526	0.32
κ-filtered MV	9.09	0.871	0.646	1.33

The original retained pool is 23% more correlated while the SNR proxy improves by 4.1×. False agreement accounts for 12.9% of Qwen's retained candidates and 40.2% of Gemma's. Gemma's retained precision nevertheless rises from 33.4% to 59.8%: false agreement need not be negligible for the likelihood-ratio condition to hold. Pool-level improvements require separate empirical verification.

K Bidirectional Selection Analysis

The exchange tables compare per-question correctness in both directions. Every count and derived win/loss ratio retains its reported value pending recomputation after excluding GPS-full. These paired tables and the older recovery-flow taxonomy must be reconciled during that audit; the recovery-only figure is not a substitute for net-change accounting.

Table K.1: Claim MV versus CCV correctness. Rows are Claim MV outcomes; columns are CCV outcomes. CCV records 237 wins and 59 losses (4.02:1).

	CCV correct	CCV incorrect
Claim MV correct	1,147	59
Claim MV incorrect	237	266

Table K.2: EAV versus CCV correctness. Rows are EAV outcomes; columns are CCV outcomes. CCV records 121 wins and 39 losses (3.10:1).

	CCV correct	CCV incorrect
EAV correct	1,263	39
EAV incorrect	121	286

In all 33 incorrect execution-fallback cases against Claim MV, no candidate executes to the correct answer; 15 exhibit unanimous 16/16 execution consensus. These failures show why a large fallback cluster cannot certify correct visual premises. The reported false-agreement losses often involve small retained groups, showing the separate risk of discarding a correct raw majority.

L Verification and Preparation Costs

Table L.1 separates verification on an existing pool from the preparation needed to obtain it. CCV needs no additional model calls after SKETCH candidates have been generated, but the complete pipeline includes teacher reformatting and generator SFT. The entries below are the available cost observations, not a complete compute ledger for all experiments or teacher calls.

M Other Matching and Scoring Constants

Besides the development-selected execution threshold and the relative-tolerance sweep in Appendix H, the reported sensitivity checks cover absolute tolerance, answer-clustering tolerance, option mapping, and the claim-cluster weight. Absolute and clustering tolerance sweeps span four

Table K.3: Reported regression categories. The available taxonomy does not exhaustively account for the EAV losses; categories and totals await reconciliation rather than assigning an unsupported residual.

Category	Against Claim MV	Against EAV
False κ agreement	26/59	30/39
Median wrong agreement-group size	3	2
Incorrect execution fallback	33	Not specified
Threshold effect	Not applicable	7

Table L.1: Available cost comparison. Pool-dependent totals are marked for recomputation. DSL execution is shared by execution-based selectors; external scorers incur separate model inference.

Component	Cost
CCV aggregation	2–4 ms/question on one CPU core; 4–7 s for the original pool
DSL execution	$<$10 s for 27,344 candidates; shared with Exec MV, EAV, and PROVE-style filtering
SKETCH candidate length	Mean approximately 86 tokens
One-time LoRA SFT	Approximately 3.5 GPU-hours to the evaluated checkpoint; approximately 10.6 GPU-hours for the full run
URSA-RM weighted MV	One reward-model forward pass per candidate
CISC	One confidence self-assessment per candidate
Teacher reformatting	Required for generator preparation; complete compute accounting pending

orders of magnitude with unchanged reported accuracy. The default option-mapping tolerance of 0.02 is not a local optimum, and claim weights in $[0, 2]$ yield little variation. Table M.1 records the available scope without inventing missing endpoints or accuracy values; those numeric results remain pending.

Table M.1: Sensitivity-table structure for the remaining constants. "Pending" denotes a numeric result absent from the supplied revision material, to be filled by the engineering audit. Red sweep results require recomputation.

Constant	Available sweep scope	Reported result / numeric accuracy
Absolute tolerance	Four orders of magnitude; endpoints pending	Unchanged / pending
Clustering tolerance	Four orders of magnitude; endpoints pending	Unchanged / pending
Option-mapping tolerance	Default 0.02; grid pending	Default not a local optimum / pending
Claim-cluster weight	$[0, 2]$; default 0.5	Small variation / pending
Agreement bonus	Common $+1$ in $\mathcal{C}_\kappa$	No effect on argmax (exact)

Why the $+1$ bonus is irrelevant to selection. For every $c_i \in \mathcal{C}_\kappa$, $\kappa_i = 1$ and the score is $w_c|\text{claim_cluster}_i| + |\text{exec_cluster}_i| + 1$. Removing the common constant preserves all score orderings and ties. When $\mathcal{C}_\kappa$ is empty, neither fallback branch uses this score. Thus the bonus cannot change the algorithm's output under a fixed tie-breaking rule.

N Notation

Table N.1: Notation for agreement, precision, dependence, and routing.

Symbol	Meaning
X_i, Y_i, z^*	Normalized neural claim, execution result, and correct answer
$\perp$	Failed execution; counted as incorrect and excluded from agreement
K, K_κ	Candidate count and retained agreement-pool size
κ_i	Claim–execution agreement indicator, excluding $\perp$
$\mathcal{C}_\kappa$	Set of candidates with $\kappa_i = 1$
t_κ, f_κ	Joint probabilities of agreement with correct and incorrect execution
p_e, p_κ	Execution precision before and after agreement filtering
a, b	Agreement probabilities conditional on correct and incorrect execution
a/b	Agreement likelihood ratio in the posterior-odds identity
ρ_e, ρ_κ	Execution-correctness correlation between within-question candidate pairs, before and after filtering
ϕ	Within-question correlation between claim and execution correctness across candidates
τ_{exec}	Minimum execution-cluster size for execution fallback
μ_i	Cross-candidate, cross-layer agreement feature used in the learned ablation
w_c	Claim-cluster weight, fixed to 0.5 in CCV

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